

T008 Validation Report — Idempotency-Key Interceptor

Task: T008 [P] Add Idempotency-Key interceptor

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Branch: feature/t008-idempotency-interceptor

Verdict: **PASS**

Speckit Pre-requisite Checks

#	Check	Result	Details
1	Feature branch	PASS	feature/t008-idempotency-interceptor
2	plan.md exists	PASS	specs/.../plan.md
3	spec.md exists	PASS	specs/.../spec.md
4	tasks.md exists	PASS	specs/.../tasks.md
5	research.md exists	PASS	specs/.../research.md
6	data-model.md exists	PASS	specs/.../data-model.md
7	quickstart.md exists	PASS	specs/.../quickstart.md
8	T002 dependency	PASS	Config + PostgreSQL complete
9	T004 dependency	PASS	Prisma ORM initialized

Implementation Validation

#	Step	Result	Details
1	npm test -- idempotency	PASS	18/18 tests passed
2	Duplicate POST prevention	PASS	First cached, second replayed
3	Non-mutation bypass	PASS	GET/OPTIONS not checked
4	No key = passthrough	PASS	Missing key = execute normally

5	Status replay	PASS	Original status preserved
6	Body replay	PASS	Original body preserved
7	Actor isolation	PASS	Scoped by actor
8	Route isolation	PASS	Scoped by route
9	Method isolation	PASS	Scoped by method
10	Concurrent safety	PASS	First-winner replayed on P2002
11	Cache write fail safe	PASS	Non-constraint error = safe
12	Lookup fail safe	PASS	Falls through to handler
13	Error propagation	PASS	Handler errors forwarded
14	PUT/PATCH support	PASS	Both handled
15	DELETE support	PASS	Cached and replayed
16	npm run build	PASS	Exit code 0
17	npx prisma validate	PASS	Schema valid

Files Changed

File	Action
backend/src/common/http/idempotency.interceptor.ts	Created
backend/test/idempotency.spec.ts	Created — 18 tests
backend/prisma/schema.prisma	Updated — IdempotencyRecord model
backend/src/app.module.ts	Updated — APP_INTERCEPTOR registration
backend/tsconfig.json	Updated — jest types
backend/.eslintrc.cjs	Updated — test/ ignorePattern

Risk Review

Risk	Status	Mitigation
Double mutation execution	PASS	Unique scoped key + DB constraint
Replay corruption	PASS	Response body/status stored exactly

Actor collision	PASS	Actor part of scoped key
Route collision	PASS	Route part of scoped key
Unsafe payload caching	PASS	SHA-256 hash stored, not raw body
Concurrent duplicates	PASS	P2002 handled — first-winner replayed
Prisma transaction issues	PASS	Graceful fallback on DB errors
Financial invariant violations	PASS	Scoped key prevents double-billing/double-assignment

Overall Verdict

PASS — All 17 checks passed + 8 risk reviews passed. T008 is valid.